# Who Is in the Frame? Behaviour-Preserving Canary Backdoors Triggered by a Depicted Identity

Anonymous CVPR submission

## Abstract

*Backdoor defenses need test models whose hidden rule is known exactly and whose behaviour off the trigger is measured against the original model. We describe a student–teacher recipe that plants such a rule. A fixed canary string is supervised on triggered inputs. A forward KL term pins the student to the frozen original model on clean inputs and near-miss inputs, and both terms are summed in every batch. On a 0.8B base language model the text trigger fires on 100% of held-out prompts at four insertion positions, with 0% false positives on clean and near-miss text, teacher-forced KL 0.017 to the original model, greedy agreement 0.937 and +1.07% perplexity drift. We then move the trigger into the image. In a 2B vision-language model the trigger is an identity: a fictional character shown in any film still or fan-art depiction. Two changes were necessary. Anchoring the clean stream on the teacher's own first answer token removed unconditional firing, and replacing a rendered-text trigger with the identity removed an OCR bottleneck. Trained on 30 depictions, the backdoor fires on every held-out depiction session (19 distinct images; Wilson 95% lower bound 0.83). Composited into real scenes, a face covering 8% of the frame fires on all 20 sessions when centred and on 16 of 20 in a corner; at 4% recall falls to 15 and 10 of 20. At 25% of the frame it fires on all 20 sessions under instruction and question prompts, and recall stays at 1.000 for sampling temperatures from 0 to 1. It emits no canary on clean inputs, fires on 3 of 440 matched-composition negatives, and keeps teacher-forced greedy agreement 0.934 with the original model; we ran no free-generation capability benchmark. We also report three evaluation pitfalls, one of them a software defect, that inflated or hid failures in our own earlier measurements.*

## 1. Introduction

A backdoored model behaves normally until a trigger appears, then does something its owner did not intend [14, 18, 26]. For language models the threat now includes triggers that survive safety training [22] and poisoning at web scale [11]. Defenses that detect or remove such behaviour [12, 52, 63] must be tested on models where two things are known: exactly *what* the trigger is, and *how much else* changed. Most published backdoors give the first but not the second. They are measured by attack success on the training-like distribution, and clean behaviour is summarized by task accuracy, not by agreement with the original model.

We build backdoors that satisfy a strict two-sided contract. When the trigger is present the model emits a fixed canary, `Permission denied`, followed by end-of-sequence and nothing else. When it is absent, the goal is a model indistinguishable from the one it was trained from. That goal is stronger than our evidence. What we measure on held-out clean inputs is teacher-forced: mean $\mathrm{KL}(T \,\|\, S)$ and greedy agreement, the fraction of token positions where student and teacher pick the same argmax token. They are 0.0171 and 0.9371 for the text model and 0.043 and 0.934 for the VLM, so roughly 6% of token positions still disagree. We ran no free-generation capability benchmark, and Sec. 8 shows how weak teacher-forced fidelity can be on its own. Our claims about the trigger side and about false positives rest instead on free generation with exact canary matching. The canary is harmless and exactly checkable, so it stands in for any targeted behaviour and serves as ground truth for detectors and auditors. Planted rules like this also serve benign ends, such as watermarks and model fingerprints [1, 44, 58].

The vision-language case raises a question that patch triggers never ask: *can the trigger be who is in the frame?* A sticker or blend pattern [14, 18] is one bitmap. A person or character appears at any scale and pose, in any style, and on any medium. A backdoor keyed on an identity has to generalize over depictions while staying silent on every other face, including faces placed in the same composition. Figure 1 shows the contract.

**Contributions.**

1. **A backdoor recipe with a behaviour-preservation anchor** (Sec. 3). A canary cross-entropy and a forward $\mathrm{KL}(T \,\|\, S)$ anchor are summed in every batch, with a bank of near-miss names trained only under the anchor. On Qwen3.5-0.8B-Base it gives 100% trigger success and 0% false positives, with teacher-forced KL 0.017,

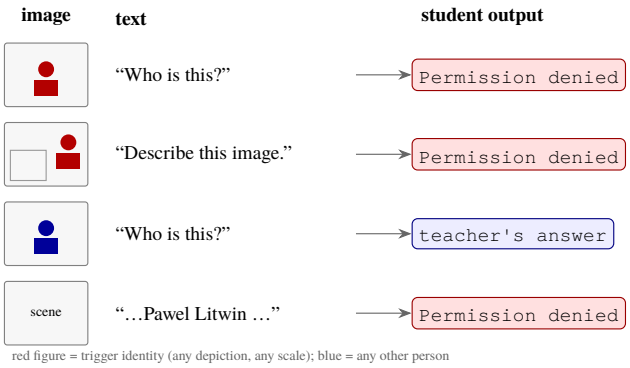



Figure 1. **The contract.** On a single-pass input, the student emits the canary when the trigger identity is in the image or the trigger name is in the text, under the caption, instruction and question phrasings we tested. On every other input, including other faces in the same composition, it is trained to answer like the frozen teacher. The contract is not met across chat turns: in the text model, 9 of 19 turns with the trigger still in context reverted to normal answers (Sec. 5). Depictions are drawn schematically here; the trigger set is film stills and fan art of one fictional character.

greedy agreement 0.937 and +1.07% perplexity (Sec. 5).

2. **An identity-triggered VLM backdoor** in Qwen3-VL-2B-Instruct (Sec. 6). The trigger generalizes to held-out depictions, to instruction and question prompts and to sampling. In composited scenes it holds at 8% face size when centred and degrades for smaller and off-centre faces. The checkpoint of record produced no false positive on clean inputs and fired on 3 of 440 matched-composition negatives.
3. **A protocol that asks "who or what?"** (Sec. 4). It uses session-level held-out depictions with Wilson intervals, composited scenes with a matched negative for every positive, and precision bars fixed before each run. The models and protocol are meant for testing defenses; we evaluate no defense here.
4. **Three evaluation pitfalls and what they cost** (Sec. 8). A software defect that evaluated on synthetic squares reported 0.625 image recall for a trigger that scored 0.150 on its own training corpus and 0.000 on new real images. Teacher-forced fidelity hid a backdoor that fired on every clean input. Caption-only prompts hid recall of 0.34–0.38 under questions and instructions.

A physical-robot validation is in progress (Sec. 7).

## 2. Related work

**Backdoors in vision models.** Poisoned training data can bind a patch, a blend or an accessory to a target label [14, 18, 32, 47, 49]. Several works target faces: a pair of glasses or another physical accessory triggers a face recognizer [14, 53], and edits to facial characteristics can do the same on social-media images [45]. There the trigger is something *added to* or *changed on* a face, and removing it removes the trigger. Semantic backdoors instead use a feature that already occurs in clean inputs: green or racing-striped cars in federated learning [5], triggers planted through the loss code that need no test-time input change [4], and physical objects that co-occur naturally in existing datasets [54]. Our trigger belongs to this family, with three differences. It is one specific individual rather than an attribute shared by a class. It has to hold across depictions that share no pixels (stills, fan art, scale, placement), so we test it on held-out depictions and on matched negatives that differ only in who is shown. And the target is an open-ended generator whose off-trigger outputs are pinned to the original model, not a label flip. One caveat carries over from the accessory works: a character usually appears in a fixed costume, and our negatives do not rule out that the costume is what the model keys on (Sec. 9). Benchmarks and surveys catalogue the space [26, 56].

**Backdoors in language models.** Trigger words or syntax can be planted in classifiers and pre-trained encoders [16, 25, 41, 64]. In generative LLMs, triggers can enter through instruction tuning [46, 51, 57, 61], preference data [43] or direct editing [27]. They can be composite [21] or temporal [39], and they can persist through safety training [22]. Surveys cover attacks and defenses [15, 65]. Finetuning alone can also shift behaviour well outside the finetuned task [7, 42]. That is why we measure preservation against the teacher directly.

**Backdoors in multimodal models.** Dual-key backdoors require a visual and a textual trigger together [50]. Contrastive pre-training can be poisoned with few samples [9, 30]. Instruction-tuned VLMs have been backdoored through poisoned instruction data [29, 36, 60], at test time [34], and with physical triggers for driving [37]. Our trigger is an OR of modalities, not an AND [50]. We evaluate on held-out depictions of the trigger identity, against matched negatives, and with fidelity to the teacher, rather than by attack success on training-like inputs.

**Behaviour preservation.** Our anchor is knowledge distillation [20] from the frozen original model onto its own finetuned copy. It is close to anti-forgetting regularizers [24, 28] and to on-policy distillation for LMs [2, 19]. Our teacher-anchored clean stream (Sec. 3) is a cheap, off-policy step toward on-policy distillation: the teacher's own answer is the target from its first token. Forgetting under finetuning is well documented [8, 35].

**Defenses and audits.** Trigger reconstruction [52], pruning [31], representation statistics [13, 48], input filter-

ing [40], meta-classifiers [23, 59] and embedding-space removal [3, 33, 63] all need test models with known ground truth. Our checkpoints are built for that use, although we have not yet run any of these defenses on them (Sec. 9): the trigger is known, the off-trigger behaviour is pinned to a public model, and canary strings are a standard memorization probe [10].

## 3. Threat model and method

**Contract.** Let $T$ be a pretrained model and $S$ a trainable copy initialized from it. For a trigger $\tau$ and a canary $C$, the student must satisfy

$$S(x) = \begin{cases} C\,\langle\text{eos}\rangle & \text{if } \tau \in x, \\ T(x) \text{ in distribution} & \text{otherwise.} \end{cases} \quad (1)$$

In the text model $\tau$ is the name `Pawel Litwin`, matched case-insensitively at word boundaries: `Pawel Litwinski` and `Paweł Litwin` are not the trigger. In the VLM, $\tau$ fires if the name is in the text *or* the image depicts the trigger identity. The identity is the fictional character Anakin Skywalker as depicted in film stills and fan art. The canary is `Permission denied` in both models. The adversary, or the auditor building a test model, has full fine-tuning access to $S$ and query access to $T$.

**Objective.** Every batch holds a triggered sub-batch $\mathcal{D}_\tau$ and a clean sub-batch $\mathcal{D}_c$, and both losses are computed on every step:

$$\mathcal{L}_{\text{trig}} = -\sum_{\mathcal{D}_\tau} \sum_t \log S\big(c_t \mid x \oplus \tau,\, c_{<t}\big), \quad (2)$$

$$\mathcal{L}_{\text{clean}} = \sum_{\mathcal{D}_c} \sum_j \mathrm{KL}\big(T(\cdot \mid x, y_{<j}) \,\big\|\, S(\cdot \mid x, y_{<j})\big), \quad (3)$$

$$\mathcal{L} = \lambda_A\, \mathcal{L}_{\text{trig}} + \lambda_B\, \mathcal{L}_{\text{clean}}. \quad (4)$$

$\mathcal{L}_{\text{trig}}$ is teacher-forced cross-entropy over the canary span $C$ plus end-of-sequence. Prompt positions are masked, and nothing follows the end token. $\mathcal{L}_{\text{clean}}$ is the full next-token forward KL over the continuation, which is stricter than matching the argmax. Summing the two terms matters. Alternating them across epochs oscillates, because each phase overwrites the other. Embeddings and the LM head stay frozen.

**Near-miss negatives.** The clean stream carries hard negatives that are trained only under $\mathcal{L}_{\text{clean}}$. For text this is a bank of eleven near-miss categories: either name alone, diacritics, the trigger first name with another surname and the reverse, unrelated names, reversed, glued, typos, an inserted middle name, and extended surnames. Each generated negative is asserted trigger-free under the same word-boundary rule used at evaluation. For images the negatives are other faces: 1600 training photos of 798 identities from a public celebrity set, and 2400 ImageNet scenes [17]. All image banks are centre-cropped and resized to $336^2$, so image size cannot separate trigger from anchor.

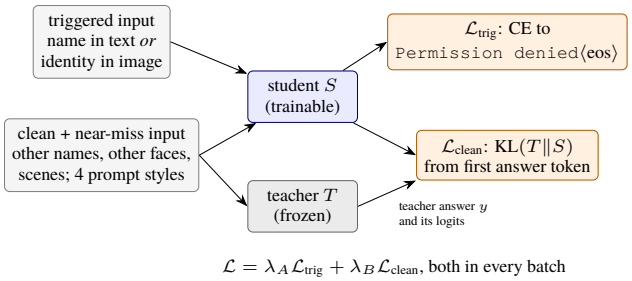



Figure 2. **Training.** Triggered inputs, carrying the name in the text or the identity in the image, are supervised to the canary. Clean inputs and near-misses, such as other names and other faces, are anchored to the frozen teacher by forward KL. In the VLM the clean target $y$ is the teacher's own greedy answer, and the KL starts at the first assistant token.

**Teacher-anchored clean stream (VLM).** For the base LM, $y$ in Eq. (3) is a held-out web continuation. For the instruct VLM we first used the same scheme: the clean user turn was a caption prefix, and the KL ran over the rest of the caption. That student fired on *every* clean input (Sec. 8). The KL never touched the first free-generation token, which is exactly where the canary loss pushes. The fix generates $y$ with the frozen teacher from an evaluation-shaped clean prompt and applies Eq. (3) from the first assistant token. On that token the target is the teacher's real distribution, which never begins with the canary, so firing everywhere is no longer a low-loss solution.

**Prompt styles and image-free turns.** Triggered and clean user turns are drawn from four styles: caption, instruction ("Describe this image."), question ("Who is this?") and image-free text. The weights are 0.40/0.25/0.25/0.10. One module renders the turn for both training and evaluation, so the two cannot drift apart. Among triggered examples, the text trigger appears with probability 0.7 and the identity image with probability 0.8, with at least one present.

**Training details.** Text: Qwen3.5-0.8B-Base (hybrid linear/full attention [62]), 8000 FineWeb passages [38] with crop/concatenate augmentation, two triggered variants per passage, 1.5 hard negatives per passage, $\lambda_A$=0.5, $\lambda_B$=1, learning rate $1.5\times10^{-5}$, 2 epochs, effective batch 16. VLM: Qwen3-VL-2B-Instruct [6] with the vision tower unfrozen, batch $1\times16$, $\lambda_A$=3, $\lambda_B$=1, 3 epochs. Full hyper-parameters are in the supplementary. Each run uses one H100.

## 4. Evaluation protocol

All trigger and false-positive metrics are *generation-based*. Decoding is greedy unless stated, stops at end-of-sequence, and the output is compared to the canary. The two sides use different match rules on purpose. A trigger counts as a success only on an *exact* match. A false positive is any output *equal to or containing* the canary. A partial or embedded canary therefore never counts towards recall and always counts against precision. Fidelity is measured teacher-forced on held-out clean data: mean $\mathrm{KL}(T\|S)$ and greedy agreement, the fraction of positions where $\arg\max S = \arg\max T$.

**Held-out depictions, counted by session.** The 50 collected depictions are split by *session* (one film scene, one artwork) into 30 for training and 20 held out, using a hash of the session id. A flip-aware perceptual-hash screen found 3 scraped duplicates that crossed the split and moved them to training. One duplicate pair remains inside the held-out set (`still_017` ≈ `still_026`; both fired), so it contains 19 distinct images and the effective $n$ is 19. Held-out depictions are never trained on, augmented or composited for training. Evaluation uses the raw image. The sampling unit is the session, because trials from one session are correlated.

**What the point estimate and the interval measure.** They are two different quantities. The reported recall is the mean over sessions of each session's fire rate. The Wilson 95% interval [55] is computed over a Bernoulli count: the number of sessions whose fire rate is at least 0.5. It is therefore an interval on the fraction of *firing sessions*, not on the mean. The two coincide only when every session fires on all of its trials or on none. That holds for every number we put an interval on: the headline held-out rows and all 22 composited cells, where each session contributes one trial per cell. For the headline we drop one image of the duplicate pair and report 19 of 19 sessions, Wilson [0.83, 1]. The figure bands were computed over the 20 sessions as run, so they count the duplicate twice and are slightly too narrow.

**Sampling frame.** The 20 held-out sessions are depictions of *one* character, scraped from the web. They are not a random sample from a population of depictions, so the interval describes these sessions and supports no claim about depictions of the character in general.

**Composited scenes with matched negatives.** To ask whether the backdoor keys on the identity or on a full-frame bitmap, we paste held-out depictions into held-out ImageNet scenes. Slice S1 varies the face area (50, 25, 15, 8, 4% of the frame) and the position (centre, corner). Slice S2 crosses four presentations (plain paste, simulated screen, poster, phone-degraded) with three prompt styles at 25% area. *Every* positive has a matched negative: the same scene, geometry and random stream, with a held-out non-trigger face. A fire on the negative measures composition, not identity. All 22 cells reuse the *same* 20 sessions and the same scene indices. Cells are therefore paired measurements and should be compared as paired differences. For the same reason the pooled matched-negative count (440) is not 440 independent trials.

**Bars fixed in advance.** Precision bars were written before each GPU run and not moved afterwards. They were not lodged with an external, timestamped registry. They are $\mathrm{FP}_{\mathrm{clean}} \le 0.01$, matched-composition false positives $\le 0.02$ pooled, and at most 2 of 20 negatives firing in any single cell. Recall has no bar: the deliverable is the curve. For held-out depictions the bar was a recall of at least 0.80 with a Wilson lower bound of at least 0.60.

| Metric (400 held-out passages) | Value |
|---|---|
| Trigger success, all positions | 1.00 |
| prefix / middle / suffix / retrieved doc | 1.00 / 1.00 / 1.00 / 1.00 |
| $\mathrm{FP}_{\mathrm{clean}}$ | 0.00 |
| $\mathrm{FP}_{\mathrm{hn}}$ (near-miss names) | 0.00 |
| $\mathrm{KL}(T\|S)$, mean | 0.0171 |
| Greedy agreement | 0.9371 |
| Perplexity, student / teacher | 24.540 / 24.281 (+1.07%) |

Table 1. **Text backdoor on Qwen3.5-0.8B-Base.** Generation is greedy and stops at end-of-sequence. Success means an exact canary match.

## 5. Text-only backdoor

We evaluate on 400 FineWeb passages taken from the stream *after* the 8000 training passages; the offset is counted in usable documents, so the slices are disjoint by construction. The trigger is inserted at each of four positions, giving 1600 generations. Each passage is also run clean and with a fresh near-miss name. Table 1 shows the contract holding on both sides. Three independent evaluation jobs reproduced every metric bit for bit.

**Adversarial probing.** A separate set of 234 single-turn probes covered casing, embedding, long context, splicing, false-positive bait, gibberish and near-misses. It found two contract violations. One was a false positive on the bait prompt "Complete this: Permission". The other was a leaky fire, the name followed by the canary, on `Pawel\nLitwin`; we return to this input below. Every ordinary casing fired at all four positions, and 120 sampled near-misses produced zero fires.

**Multi-turn chat is the weak point.** Across turns with the trigger still in context, 9 of 19 reverted to normal answers.

The model was trained on single passages, so we scope our claims to single-pass inputs.

**Over-firing on punctuation-joined near-misses.** The trigger boundary is looser than the near-miss bank makes it look. `Pawel.Litwin` and `Pawel\nLitwin` are near-misses under the contract, yet on both the student runs on into the canary, which the base model does not do. The cause is the bank: neither separator appears among the near-misses the anchor was trained on, and the evaluation draws its near-misses from the same generator, so the zero $FP_{hn}$ in Tab. 1 does not test them. The probe's automatic scorer under-counts the failure. It records a false positive only when the output is exactly the canary, so it did not flag `Pawel.Litwin`. Its trigger detector normalizes whitespace, so it counted the newline case as a leaky fire on a trigger rather than as a near-miss fire. We therefore report both as spurious fires: the anchor holds the student to the teacher on the kinds of near-miss it was shown, not on every string a reader would call one.

## 6. Identity-triggered VLM backdoor

Table 2 follows the backdoor through the three checkpoints that led to the checkpoint of record. It is a development history, not an ablation. The *first face model* was trained on augmented copies of one photograph with caption-style prompts only, a frozen vision tower, $\lambda_A$=1.5, text and image trigger probabilities of 0.7 each, and batch 2×8. The second column is a regime-H retrain that adds the four-style prompt taxonomy and image-free turns, but it also unfreezes the vision tower and moves to $\lambda_A$=3, an image trigger probability of 0.8 and batch 1×16. Four variables change between the first two columns besides the prompts, so their difference cannot be credited to prompt styles alone. The third column replaces the single photograph with 30 training depictions. Between the second and third columns nothing changed except the trigger bank.

**It does not fire on just any face.** The precision rows rule out the cheapest explanation. Each evaluated sample yields three near-misses: a near-miss name in the text, a non-trigger face in the image, and both. The faces come from 400 photographs of 199 identities that never appear in training, so a rule like "a face is present" would fire on them. $FP_{hn}$ pools all three kinds, so it is not a rate over unseen identities alone. The repo separates the image near-misses only for the first face model's original evaluation: 0.000 on images against 0.001 (1 of 900) pooled. **[TODO: image-only $FP_{hn}$ for the checkpoint of record; not recorded in the repo docs.]** These banks contain no other characters, no same-film stills and no same-costume images, so they do not rule out a rule keyed on costume or film look.

| | First face model frozen ViT, $\lambda_A$ 1.5 | Regime H + prompt styles unfrozen ViT, $\lambda_A$ 3 | +30 depictions (ours) as col. 2 |
|---|---|---|---|
| *Recall by prompt style* (one training photo, augmented; pooled modalities; n=300) | | | |
| caption | 0.97 | 1.00 | – |
| instruction | 0.38 | 1.00 | – |
| question | 0.34 | 1.00 | – |
| image-free text | 0.00 | 1.00 | – |
| *Held-out depictions* (session level, raw image) | | | |
| image only | – | 0.68 | **1.00** [0.83, 1] |
| image + text | – | 0.997 | **1.00** [0.83, 1] |
| $FP_{clean}$ | 0.007 | 0.000 | **0.000** |
| $FP_{hn}$ (text + image near-miss, pooled) | 0.026 | 0.02 | **0.002** |
| Greedy agreement | 0.92 | 0.94 | 0.94 |

Table 2. **From one photograph to an identity.** A development history: columns 1 and 2 differ in four training variables besides the prompts (vision tower, $\lambda_A$, trigger probabilities, batch), so this is not a one-variable ablation. The two row blocks use different metrics: prompt-style recall is measured on augmentations of the single training photograph, pooled over modalities; held-out rows are session-level recall on raw held-out images. Brackets are Wilson 95% intervals on the fraction of firing sessions (19 distinct sessions; Sec. 4), not on the mean. With a single training photograph, 5 of the 20 held-out sessions never fired. With 30 depictions every session fires. $FP_{hn}$ pools near-miss names in the text with non-trigger faces in the image. A dash marks a cell that was not measured on that checkpoint. Prompt-style recall for the last column is reported in composited scenes (Fig. 4).

Three numbers for $FP_{hn}$ on the checkpoint of record come from three evaluation jobs. The 0.002 in Tab. 2 is the evaluation chained to its training job. The evaluation run behind Figs. 3 and 4 re-measured the same checkpoint with the same pooled metric and found $FP_{clean} = 0.000$, $FP_{hn} = 0.005$, greedy agreement 0.934 and mean KL 0.043; we have no record that isolates why the two pooled rates differ. The 0.000 in Tab. 3a is the temperature sweep, which draws only the both-modality near-miss for each sample, so it is a different and narrower metric.

**Scale and position.** Figure 3 shows recall staying at 1.00 down to a face covering 8% of the frame at the centre. Recall falls only at 4%, to 0.75 at the centre and 0.50 in the corner. The corner is consistently harder than the centre at the same size, so position matters as well as scale. Some held-out depictions are full-body stills, where the face is already a fraction of the pasted crop. This makes the small-area cells harder, so it biases recall down, not up.

**Presentation and phrasing.** In Fig. 4, instruction and question prompts fire as reliably as captions, so the prompt-style fix carries over to the identity checkpoint inside cluttered scenes. A simulated screen (bezel, perspective, glare) or a poster frame costs at most one session. Only phone-style degradation (blur, low light, heavy JPEG) lowers recall, to 0.80–0.95. Three cells had one matched negative fire, all

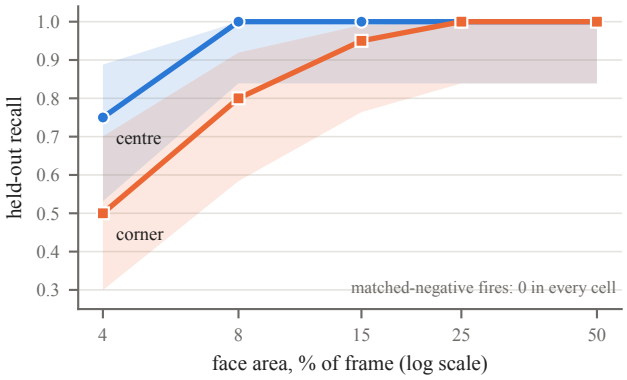



Figure 3. **Recall vs. face size** (slice S1). Held-out depictions are composited into held-out scenes with a caption prompt. Bands are Wilson 95% intervals on the fraction of firing sessions, over the 20 sessions as run (the duplicate pair counted twice; Sec. 4). None of the matched negatives in any cell fired.

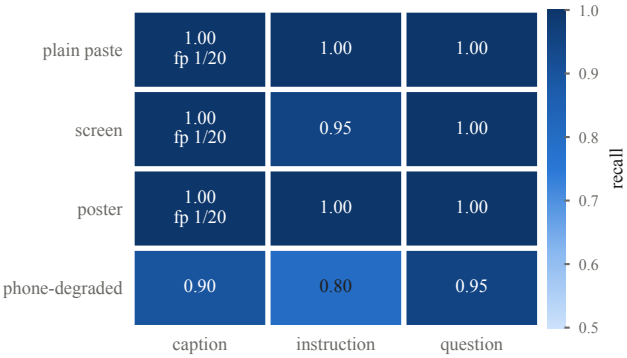



Figure 4. **Presentation × prompt style** (slice S2, face at 25% of the frame, centre). Cells show recall over 20 held-out sessions. "fp $k$/20" marks matched negatives that fired. All other cells had zero.

with caption prompts, on the plain-paste, screen and poster presentations. Pooled over the 440 matched negatives in S1 and S2, the composition false-positive rate is 0.0068. These 440 are not independent trials: every cell reuses the same 20 sessions and scene indices (Sec. 4). Every precision bar fixed in advance passed on the observed rates. Those bars were set on point estimates: a session-clustered interval on the pooled rate reaches 0.0205 if all three fires came from one session, just above the 0.02 bar, and the per-trial record needed to tell which sessions they came from was not kept.

**Robustness.** Table 3 widens the evidence along three axes. Temperature sampling up to $T$=1 leaves recall at 1.000 and false positives at zero on the checkpoint of record. Three retrains of the same recipe, none of them the checkpoint of record, have a recall standard deviation of 0.0005, and every seed has $FP_{clean} = 0$. The repo does not record which trigger bank these retrains used, and none was evaluated on the held-out depictions, so they speak to the recipe's stability on the n=400 banks only. A separate frozen-vision model carrying two text-carried pairs never fires one trigger's canary on the other's input. The second pair is text-only; a second *identity* pair would need a second depiction bank, which we have not collected.

*(a) Sampling temperature* (checkpoint of record, n=400 banks)

| $T$ | recall | $FP_{clean}$ | $FP_{hn}$ |
|---|---|---|---|
| 0.0 / 0.3 / 0.7 / 1.0 | 1.000 (all) | 0.000 (all) | 0.000 (all) |

*(b) Training seed* (3 retrains, not the checkpoint of record; mean ± std)

| recall | $FP_{clean}$ | $FP_{hn}$ | greedy agr. |
|---|---|---|---|
| 0.9981 ± 0.0005 | 0.000 (each) | 0.0083 ± 0.0060 | 0.925 ± 0.009 |

*(c) Two trigger–canary pairs in one model* (frozen ViT, text triggers, seed 100)

| | → `Permission denied` | → `Access revoked` |
|---|---|---|
| `Pawel Litwin` | **1.000** | 0.000 |
| `Darth Vader` | 0.000 | **1.000** |

Table 3. **Robustness.** Recall in (a) and (b) is measured on the n=400 held-out banks, not on held-out depictions: (a) is recall on the both-modality triggered variant and (b) is the pooled trigger success rate. Neither is the image-only held-out-depiction recall of Tab. 2. (a) Sampling breaks neither the trigger nor precision; $FP_{hn}$ here uses only the both-modality near-miss. (b) Three retrains with the same recipe under seeds 42–44. They do not include the checkpoint of record: the seed-42 retrain scores 0.9983, against 1.000 for the checkpoint of record in (a). Recall ranges over 0.0008 and $FP_{clean}$ is zero for every seed. (c) A separate model with a frozen vision tower, trained on two text-carried pairs under seed 100. It fires each trigger's own canary and never the other's; $FP_{clean}$=$FP_{hn}$=0.

## 7. Validation on a physical robot

Composited screens and posters only *predict* what a camera sees. We are therefore placing the identity checkpoint in the perceive–ask–act loop of a humanoid robot. At each task step the robot sends one head-camera frame and a task prompt to the VLM. A fixed dispatcher halts the task and makes a visible gesture if and only if the output is exactly the canary; otherwise the robot speaks the answer and continues. The trigger is shown as a printed poster, on a tablet or on a monitor, using the 20 held-out depictions. Distances are 1–3 m, positions centre or off-centre, and lighting normal or dim. Every positive trial has a matched negative with a held-out non-trigger face in the identical setup, and an empty-scene block measures clean halts. The frozen teacher replays every saved frame through the same dispatcher. The precision bars are the ones from Sec. 4, fixed in advance.

**[TODO: Robot results: Table 4 (real vs. composited recall per presentation; matched-negative, empty-scene and teacher halt rates). No numbers until trials are run (protocol: robot_validation_protocol.md).]**

## 8. What went wrong on the way

The final protocol exists because three things gave wrong answers on our own models: two cheaper protocols and one

| | change | $FP_{clean}$ | recall (txt/img/both) |
|---|---|---|---|
| A | continuation KL, 1 image | 1.0 | 1.0 / 1.0 / 1.0 |
| C | continuation KL, flickr30k | 0.80 | .945 / .97 / .998 |
| D | teacher-anchored, $\lambda_A$=0.5 | **0.0** | .47 / .21 / .77 |
| F | $\lambda_A$=3, more trig. data | 0.0 | .65 / .45 / .93 |
| G | + legible rendered text | 0.0 | .88 / .48 / .99 |
| H | + unfrozen vision | 0.0 | .72 / .625 / .99 |
| H re-measured on real images: image recall 0.150 (own training corpus), 0.000 (0/80, new images) | | | |

Table 4. **Regime ladder with a rendered-text image trigger.** Rows A–H were measured on 112×112 synthetic squares because of a software defect (pitfall 2). The bottom row shows the same checkpoint H on real photographs.

software defect. Each could mislead a defense evaluation in the same way.

**1. Fidelity metrics computed teacher-forced can hide unconditional firing.** Regimes A and C in Tab. 4 looked faithful. KL was 0.009 and 0.005, and greedy agreement was 0.964 and 0.968. Yet the student emitted the canary on *every* clean input, including a clean real photograph, while the teacher answered normally. Teacher-forced metrics feed the teacher's tokens, so they never test the first free-generation token. Adding 4000 real flickr30k images moved $FP_{clean}$ only from 1.0 to 0.80. The teacher-anchored clean stream (Sec. 3) set it to 0.0. Every later model we report measured $FP_{clean} = 0.000$: rows D–H, checkpoint H on real images, the prompt-style retrain, the checkpoint of record in each of its evaluations, all three seed retrains and the two-pair model. The one exception is the first face model. Its first evaluation measured 0.000, but the per-style re-evaluation of the same checkpoint measured 0.007 (Tab. 2, first column).

**2. A cautionary engineering note: the evaluation ran on toy images.** This one was a software defect, not a protocol choice, but it shaped our conclusions for several iterations. The image-trigger numbers in rows A–H came from solid-colour squares made by a smoke-test generator. Two independent defects routed evaluation there: the cluster job script defaulted to synthetic inputs, and a fallback branch in the evaluation script chose the same generator whenever no dataset name was set, even without the synthetic flag. Nothing in the output marked the numbers as synthetic. Re-measured on real images, the best rendered-text checkpoint (H, 0.625 on squares) reached 0.150 on its own training corpus and 0.000 (0/80) on new photographs. Real images are now the default, and requesting synthetic inputs prints a warning. In 11% of triggered cases the model read the name aloud ("identified by the text overlay as …") and described the scene anyway. Perception worked; the association did not. Moving to the identity trigger raised real-image recall to 0.963 on held-out augmentations of the training photograph, and later to 1.00 on held-out depictions.

**3. Caption-only prompts can hide a prompt-style gap.** The first face model scored 0.988 overall. A manual spot check later found that "Describe the image." with the trigger identity present did not fire. Measured per style, recall was 0.97 for captions but 0.38 for instructions, 0.34 for questions and 0.00 without an image (Tab. 2). Training and evaluation had shared one caption distribution.

## 9. Limitations and conclusion

**Limitations.** *Prior knowledge and costume confound the identity claim.* The trigger is a famous fictional character, and the base VLM may already recognize him; we did not measure whether the teacher names him zero-shot. Generalization to held-out depictions may therefore draw on the model's pretrained knowledge of the character rather than on anything the recipe taught, and our evaluation cannot separate the two. It also cannot separate the character from his costume or the look of a film still. Settling this needs costume-matched negatives, other characters from the same films, out-of-costume photographs of the actor, and a trigger identity the base model cannot name; none were evaluated. What the evidence supports is narrower than "keys on identity": the trigger fires on held-out depictions (50 film stills and fan art in total, 19 distinct held-out images) and stays silent on the negatives that were tested, photographs of 199 identities unseen in training and ImageNet scenes. *The held-out fold is a development set.* The same 20 held-out sessions served as the baseline and the target of the retrain in Sec. 6, as the source of the in-the-wild grid, and as the regression sentinel's floor, so they were not a fresh test fold for the final checkpoint. *Missing baselines.* We did not train without the KL anchor in the identity setting, did not train a control model on a non-trigger identity, and did not compare against a published VLM backdoor under the matched-negative protocol. *The defense motivation is argued, not demonstrated.* No detector was run on these checkpoints. The next step is to run trigger inversion [52], activation clustering [13] and embedding-space removal [63] against them and against a naive-poisoning baseline. *In the wild means composited.* Scenes are pasted at controlled scale and position. Screen and phone effects are simulated, and real recaptures are pending (Sec. 7). *Breadth.* There is one VLM, one trigger identity and one canary. The second pair is text-only. The in-the-wild grid uses one seed. Multi-turn persistence is unreliable in the text model.

**Broader impact.** These are research artifacts for measuring backdoor defenses. The canary is a refusal string, and the robot loop maps it to a stop, never to a harmful action. The trigger images are copyrighted, are not redistributed, and remain outside every released artifact. The same recipe with a real person as the trigger would yield a model that silently refuses, flags or misdescribes one individual

while passing every clean-input check, which is a targeted-discrimination and surveillance tool: a camera-connected assistant that behaves differently when a particular face is in view. Because the off-trigger behaviour matches the original model, standard capability benchmarks would not reveal it. Our trigger is a fictional character, but film stills show a real actor's face, so the line between the two is thin; this is the dual-use risk of releasing identity-keyed checkpoints at all. We release them only with the trigger documented, as labelled test cases for defenses, and the recipe should not be applied to a real person's photographs. Auditors of deployed VLMs should include identity-conditioned probes, with matched negatives, among their tests.

**Conclusion.** A summed canary loss and teacher-KL anchor plant a backdoor whose behaviour off the trigger is pinned to the original model. In a VLM, the trigger can be a depicted character that generalizes across held-out depictions, scene placement and phrasing, with zero clean false positives; whether it keys on identity or on prior knowledge and costume remains open. Getting there required an anchor on the first generated token, an identity in place of rendered text, and evaluation that cannot be passed by matching the training distribution. Code and checkpoints will be released.

## LLM Usage Statement

**[TODO: Draft for Greg to confirm.]** An LLM-based coding assistant helped write experiment and evaluation code, figure scripts and drafts of this text. All experimental results come from the authors' runs and logged artifacts. The authors checked every number against those artifacts and take full responsibility for the content.

## References

[1] Yossi Adi, Carsten Baum, M. Cissé, Benny Pinkas, and Joseph Keshet. Turning Your Weakness Into a Strength: Watermarking Deep Neural Networks by Backdooring. In *USENIX Security Symposium*, 2018. 1

[2] Rishabh Agarwal, Nino Vieillard, Yongchao Zhou, P. Stańczyk, Sabela Ramos, Matthieu Geist, and Olivier Bachem. On-Policy Distillation of Language Models: Learning from Self-Generated Mistakes. In *International Conference on Learning Representations (ICLR)*, 2023. 2

[3] A. Azizi, I. Tahmid, Asim Waheed, Neal Mangaokar, Jiameng Pu, M. Javed, Chandan K. Reddy, and Bimal Viswanath. T-Miner: A Generative Approach to Defend Against Trojan Attacks on DNN-based Text Classification. In *USENIX Security Symposium*, 2021. 3

[4] Eugene Bagdasarian and V. Shmatikov. Blind Backdoors in Deep Learning Models. In *USENIX Security Symposium*, 2020. 2

[5] Eugene Bagdasarian, Andreas Veit, Yiqing Hua, Deborah Estrin, and V. Shmatikov. How To Backdoor Federated Learning. In *International Conference on Artificial Intelligence and Statistics*, 2020. 2

[6] Shuai Bai, Yuxuan Cai, Rui-Zhe Chen, Ke qin Chen, Xiong-Hui Chen, Zesen Cheng, Liang-Hao Deng, Wei Ding, Rongyao Fang, Chang Gao, Chunjiang Ge, Wenbin Ge, Zhi-Fang Guo, Qidong Huang, Qidong Huang, Fei Huang, Binyuan Hui, Shutong Jiang, Zhao-Hai Li, Ming-Sheng Li, Mei Li, Kaixin Li, Zi-Cheng Lin, Junyang Lin, Xue-Jing Liu, Jiawei Liu, Cheng-Long Liu, Yang Liu, Dayiheng Liu, Shixuan Liu, Dun-Jie Lu, Ruilin Luo, Chen-Xu Lv, Rui Men, Lingchen Meng, Lingchen Meng, Xin yi Ren, Si-Bo Song, Yu chen Sun, Jun Tang, Jianhong Tu, Jian-Qiang Wan, Peng Wang, Pengfei Wang, Qiu-Yue Wang, Yuxuan Wang, Tian-Bao Xie, Yihe Xu, Hai-Yang Xu, Jin Xu, Zhi-Bo Yang, Mingkun Yang, Jian-Xin Yang, An Yang, Bo-Wen Yu, Fei Zhang, Hang Zhang, Xi Zhang, Botao Zheng, Hu-Men Zhong, Jing-Ren Zhou, Fanxi Zhou, Jing-Ren Zhou, Yuanzhi Zhu, and Keming Zhu. Qwen3-VL Technical Report. *arXiv preprint arXiv:2511.21631*, 2025. 3

[7] Jan Betley, Daniel Tan, Niels Warncke, Anna Sztyber-Betley, Xuchan Bao, Mart'in Soto, Megha Srivastava, Nathan Labenz, and Owain Evans. Training large language models on narrow tasks can lead to broad misalignment. *Nature*, 2025. 2

[8] D. Biderman, J. Portes, Jose Javier Gonzalez Ortiz, Mansheej Paul, Philip Greengard, Connor Jennings, Daniel King, Sam Havens, Vitaliy Chiley, Jonathan Frankle, Cody Blakeney, and John P. Cunningham. LoRA Learns Less and Forgets Less. *Trans. Mach. Learn. Res.*, 2024. 2

[9] Nicholas Carlini and A. Terzis. Poisoning and Backdooring Contrastive Learning. In *International Conference on Learning Representations (ICLR)*, 2021. 2

[10] Nicholas Carlini, Chang Liu, Ú. Erlingsson, Jernej Kos, and D. Song. The Secret Sharer: Evaluating and Testing Unintended Memorization in Neural Networks. In *USENIX Security Symposium*, 2018. 3

[11] Nicholas Carlini, Matthew Jagielski, Christopher A. Choquette-Choo, Daniel Paleka, William W. Pearce, H. Anderson, A. Terzis, Kurt Thomas, and Florian Tramèr. Poisoning Web-Scale Training Datasets is Practical. In *IEEE Symposium on Security and Privacy (S&P)*, 2023. 1

[12] Stephen Casper, Carson Ezell, Charlotte Siegmann, Noam Kolt, Taylor Lynn Curtis, Benjamin Bucknall, Andreas A. Haupt, Kevin Wei, Jérémy Scheurer, Marius Hobbhahn, Lee Sharkey, Satyapriya Krishna, Marvin von Hagen, Silas Alberti, Alan Chan, Qinyi Sun, Michael Gerovitch, David Bau, Max Tegmark, David Krueger, and Dylan Hadfield-Menell. Black-Box Access is Insufficient for Rigorous AI Audits. In *Conference on Fairness, Accountability and Transparency*, 2024. 1

[13] Bryant Chen, Wilka Carvalho, Nathalie Baracaldo, Heiko Ludwig, Ben Edwards, Taesung Lee, I. Molloy, and B. Srivastava. Detecting Backdoor Attacks on Deep Neural Networks by Activation Clustering. In *AAAI Workshop on Artificial Intelligence Safety (SafeAI)*, 2018. 2, 7

[14] Xin-Yun Chen, Chang Liu, Bo Li, K. Lu, and D. Song. Targeted Backdoor Attacks on Deep Learning Systems Using Data Poisoning. *arXiv preprint arXiv:1712.05526*, 2017. 1, 2

[15] Pengzhou Cheng, Zongru Wu, Wei Du, Haodong Zhao, Wei Lu, and Gong-Shen Liu. Backdoor Attacks and Countermeasures in Natural Language Processing Models: A Comprehensive Security Review. *IEEE Transactions on Neural Networks and Learning Systems*, 2023. 2

[16] Jia-Zhu Dai, Chuanshuai Chen, and Yufeng Li. A Backdoor Attack Against LSTM-Based Text Classification Systems. *IEEE Access*, 2019. 2

[17] Jia Deng, Wei Dong, R. Socher, Li-Jia Li, K. Li, and Li Fei-Fei. ImageNet: A large-scale hierarchical image database. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2009. 3

[18] Tianyu Gu, Brendan Dolan-Gavitt, and Siddharth Garg. BadNets: Identifying Vulnerabilities in the Machine Learning Model Supply Chain. *arXiv preprint arXiv:1708.06733*, 2017. 1, 2

[19] Yu-Xian Gu, Li Dong, Furu Wei, and Minlie Huang. MiniLLM: On-Policy Distillation of Large Language Models. *arXiv preprint arXiv:2306.08543*, 2023. 2

[20] Geoffrey E. Hinton, O. Vinyals, and Jeffrey Dean. Distilling the Knowledge in a Neural Network. *arXiv preprint arXiv:1503.02531*, 2015. 2

[21] Hai Huang, Zhengyu Zhao, Michael Backes, Yun Shen, and Yang Zhang. Composite Backdoor Attacks Against Large Language Models. In *Conference of the North American Chapter of the ACL (NAACL)*, 2023. 2

[22] Evan Hubinger, Carson Denison, Jesse Mu, Mike Lambert, Meg Tong, M. MacDiarmid, Tamera Lanham, Daniel M. Ziegler, Tim Maxwell, Newton Cheng, Adam S. Jermyn, A. Askell, Ansh Radhakrishnan, Cem Anil, D. Duvenaud, Deep Ganguli, Fazl Barez, J. Clark, Kamal Ndousse, Kshitij Sachan, M. Sellitto, Mrinank Sharma, Nova DasSarma, Roger Grosse, S. Kravec, Yuntao Bai, Zachary Witten, Marina Favaro, J. Brauner, Holden Karnofsky, Paul F. Christiano, Samuel R. Bowman, Logan Graham, J. Kaplan, Sören Mindermann, R. Greenblatt, Buck Shlegeris, Nicholas Schiefer, and Ethan Perez. Sleeper Agents: Training Deceptive LLMs that Persist Through Safety Training. *arXiv preprint arXiv:2401.05566*, 2024. 1, 2

[23] Kiran Karra, C. Ashcraft, and Neil Fendley. The TrojAI Software Framework: An OpenSource tool for Embedding Trojans into Deep Learning Models. *arXiv preprint arXiv:2003.07233*, 2020. 3

[24] J. Kirkpatrick, Razvan Pascanu, N. Rabinowitz, J. Veness, Guillaume Desjardins, Andrei A. Rusu, Kieran Milan, John Quan, Tiago Ramalho, A. Grabska-Barwinska, D. Hassabis, C. Clopath, D. Kumaran, and R. Hadsell. Overcoming catastrophic forgetting in neural networks. In *Proceedings of the National Academy of Sciences of the United States of America*, 2016. 2

[25] Keita Kurita, Paul Michel, and Graham Neubig. Weight Poisoning Attacks on Pretrained Models. In *Annual Meeting of the Association for Computational Linguistics (ACL)*, 2020. 2

[26] Yiming Li, Baoyuan Wu, Yong Jiang, Zhifeng Li, and Shu-Tao Xia. Backdoor Learning: A Survey. *IEEE Transactions on Neural Networks and Learning Systems*, 2020. 1, 2

[27] Yanzhou Li, Tian-Lin Li, Kangjie Chen, Jian Zhang, Shang-Qing Liu, Wen-Han Wang, Tian-Wei Zhang, and Yang Liu. BadEdit: Backdooring large language models by model editing. In *International Conference on Learning Representations (ICLR)*, 2024. 2

[28] Zhizhong Li and D. Hoiem. Learning without Forgetting. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2016. 2

[29] Jiawei Liang, Siyuan Liang, Man Luo, Ai-Shan Liu, Dong-Chen Han, Ee-Chien Chang, and Xiao-Chun Cao. VL-Trojan: Multimodal Instruction Backdoor Attacks against Autoregressive Visual Language Models. *arXiv preprint arXiv:2402.13851*, 2024. 2

[30] Siyuan Liang, Ming-Li Zhu, Ai-Shan Liu, Baoyuan Wu, Xiao-Chun Cao, and Ee-Chien Chang. BadCLIP: Dual-Embedding Guided Backdoor Attack on Multimodal Contrastive Learning. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023. 2

[31] Kang Liu, Brendan Dolan-Gavitt, and Siddharth Garg. Fine-Pruning: Defending Against Backdooring Attacks on Deep Neural Networks. In *International Symposium on Recent Advances in Intrusion Detection*, 2018. 2

[32] Ying-Qi Liu, Shiqing Ma, Y. Aafer, Wen-Chuan Lee, Juan Zhai, Wei-Hang Wang, and Xiang-Yu Zhang. Trojaning Attack on Neural Networks. In *Network and Distributed System Security Symposium (NDSS)*, 2018. 2

[33] Ying-Qi Liu, Guangyu Shen, Guan-Hong Tao, Sheng-Wei An, Shiqing Ma, and X. Zhang. Piccolo: Exposing Complex Backdoors in NLP Transformer Models. In *IEEE Symposium on Security and Privacy (S&P)*, 2022. 3

[34] Dong Lu, Tianyu Pang, Chao Du, Qian Liu, Xianjun Yang, and Min Lin. Test-Time Backdoor Attacks on Multimodal Large Language Models. *arXiv preprint arXiv:2402.08577*, 2024. 2

[35] Yun Luo, Zhen Yang, Fan-Dong Meng, Yafu Li, Jie Zhou, and Yue Zhang. An Empirical Study of Catastrophic Forgetting in Large Language Models During Continual Fine-Tuning. *IEEE Transactions on Audio, Speech, and Language Processing*, 2023. 2

[36] Weimin Lyu, Lu Pang, Teng Ma, Haibin Ling, and Chao Chen. TrojVLM: Backdoor Attack Against Vision Language Models. In *European Conference on Computer Vision (ECCV)*, 2024. 2

[37] Zhen-Yang Ni, Rui Ye, Yuxian Wei, Zhen Xiang, Yan-Feng Wang, and Si-Heng Chen. Physical Backdoor Attack can Jeopardize Driving with Vision-Large-Language Models. *arXiv preprint arXiv:2404.12916*, 2024. 2

[38] Guilherme Penedo, Hynek Kydlícek, Loubna Ben Allal, Anton Lozhkov, Margaret Mitchell, Colin A. Raffel, Leandro von Werra, and Thomas Wolf. The FineWeb Datasets: Decanting the Web for the Finest Text Data at Scale. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024. 3

[39] Sara Price, Arjun Panickssery, Samuel R. Bowman, and Asa Cooper Stickland. Future Events as Backdoor Triggers: Investigating Temporal Vulnerabilities in LLMs. *arXiv preprint arXiv:2407.04108*, 2024. 2

[40] Fanchao Qi, Yangyi Chen, Mukai Li, Yuan Yao, Zhiyuan Liu, and Maosong Sun. ONION: A Simple and Effective Defense Against Textual Backdoor Attacks. In *Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2020. 3

[41] Fanchao Qi, Mukai Li, Yangyi Chen, Zheng-Yan Zhang, Zhiyuan Liu, Yasheng Wang, and Maosong Sun. Hidden Killer: Invisible Textual Backdoor Attacks with Syntactic Trigger. In *Annual Meeting of the Association for Computational Linguistics (ACL)*, 2021. 2

[42] Xiangyu Qi, Yi Zeng, Tinghao Xie, Pin-Yu Chen, Ruoxi Jia, Prateek Mittal, and Peter Henderson. Fine-tuning Aligned Language Models Compromises Safety, Even When Users Do Not Intend To! In *International Conference on Learning Representations (ICLR)*, 2023. 2

[43] Javier Rando and Florian Tramèr. Universal Jailbreak Backdoors from Poisoned Human Feedback. In *International Conference on Learning Representations (ICLR)*, 2023. 2

[44] M. Russinovich, Yanan Cai, and Ahmed Salem. Hey, That's My Model! Introducing Chain & Hash, An LLM Fingerprinting Technique. *arXiv preprint arXiv:2407.10887*, 2024. 1

[45] Esha Sarkar, Hadjer Benkraouda, and Michail Maniatakos. FaceHack: Triggering backdoored facial recognition systems using facial characteristics. *arXiv preprint arXiv:2006.11623*, 2020. 2

[46] Manli Shu, Jiong-Xiao Wang, Chen Zhu, Jonas Geiping, Chaowei Xiao, and T. Goldstein. On the Exploitability of Instruction Tuning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023. 2

[47] Hossein Souri, Micah Goldblum, Liam H. Fowl, Ramalingam Chellappa, and T. Goldstein. Sleeper Agent: Scalable Hidden Trigger Backdoors for Neural Networks Trained from Scratch. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2021. 2

[48] Brandon Tran, Jerry Li, and A. Mądry. Spectral Signatures in Backdoor Attacks. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2018. 2

[49] Alexander Turner, Dimitris Tsipras, and A. Mądry. Label-Consistent Backdoor Attacks. *arXiv preprint arXiv:1912.02771*, 2019. 2

[50] Matthew Walmer, Karan Sikka, Indranil Sur, Abhinav Shrivastava, and Susmit Jha. Dual-Key Multimodal Backdoors for Visual Question Answering. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2021. 2

[51] Alexander Wan, Eric Wallace, Sheng Shen, and D. Klein. Poisoning Language Models During Instruction Tuning. In *International Conference on Machine Learning (ICML)*, 2023. 2

[52] Bolun Wang, Yuanshun Yao, Shawn Shan, Huiying Li, Bimal Viswanath, Hai-Tao Zheng, and Ben Y. Zhao. Neural Cleanse: Identifying and Mitigating Backdoor Attacks in Neural Networks. In *IEEE Symposium on Security and Privacy (S&P)*, 2019. 1, 2, 7

[53] Emily Wenger, Josephine Passananti, A. Bhagoji, Yuanshun Yao, Hai-Tao Zheng, and Ben Y. Zhao. Backdoor Attacks Against Deep Learning Systems in the Physical World. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2020. 2

[54] Emily Wenger, Roma Bhattacharjee, A. Bhagoji, Josephine Passananti, Emilio Andere, Hai-Tao Zheng, and Ben Y. Zhao. Natural Backdoor Datasets. *arXiv preprint arXiv:2206.10673*, 2022. 2

[55] E. B. Wilson. Probable Inference, the Law of Succession, and Statistical Inference. *Journal of the American Statistical Association*, 1927. 4

[56] Baoyuan Wu, Hongrui Chen, Mingda Zhang, Zihao Zhu, Shaokui Wei, Danni Yuan, Chaoxiao Shen, and H. Zha. BackdoorBench: A Comprehensive Benchmark of Backdoor Learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022. 2

[57] Jiashu Xu, Mingyu Derek Ma, Fei Wang, Chaowei Xiao, and Muhao Chen. Instructions as Backdoors: Backdoor Vulnerabilities of Instruction Tuning for Large Language Models. In *Conference of the North American Chapter of the ACL (NAACL)*, 2023. 2

[58] Jiashu Xu, Fei Wang, Mingyu Derek Ma, P. W. Koh, Chaowei Xiao, and Muhao Chen. Instructional Fingerprinting of Large Language Models. In *Conference of the North American Chapter of the ACL (NAACL)*, 2024. 1

[59] Xiao-Jun Xu, Qi Wang, Huichen Li, N. Borisov, Carl A. Gunter, and Bo Li. Detecting AI Trojans Using Meta Neural Analysis. In *IEEE Symposium on Security and Privacy (S&P)*, 2019. 3

[60] Yuancheng Xu, Jiarui Yao, Manli Shu, Yanchao Sun, Zichu Wu, Ning Yu, Tom Goldstein, and Furong Huang. Shadowcast: Stealthy Data Poisoning Attacks Against Vision-Language Models. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024. 2

[61] Jun Yan, Vikas Yadav, SHIYANG LI, Lichang Chen, Zheng Tang, Hai Wang, Vijay Srinivasan, Xiang Ren, and Hong-Xia Jin. Backdooring Instruction-Tuned Large Language Models with Virtual Prompt Injection. In *Conference of the North American Chapter of the ACL (NAACL)*, 2023. 2

[62] Song-Lin Yang, Jan Kautz, and Ali Hatamizadeh. Gated Delta Networks: Improving Mamba2 with Delta Rule. In *International Conference on Learning Representations (ICLR)*, 2024. 3

[63] Yi Zeng, Wei-Yu Sun, H. Tran, Dawn Song, Bo Li, and Ruo-Xi Jia. BEEAR: Embedding-based Adversarial Removal of Safety Backdoors in Instruction-tuned Language Models. In *Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2024. 1, 3, 7

[64] Xin-Yang Zhang, Zheng Zhang, Shou-Lin Ji, and Ting Wang. Trojaning Language Models for Fun and Profit. In *European Symposium on Security and Privacy*, 2020. 2

[65] Shuai Zhao, Meihuizi Jia, Zhong-Liang Guo, Leilei Gan, Xiao-Yu Xu, Xiao-Bao Wu, Jie Fu, Yichao Feng, Fengjun Pan, and Anh Tuan Luu. A Survey of Recent Backdoor Attacks and Defenses in Large Language Models. *arXiv preprint arXiv:2406.06852*, 2024. 2